Exponential Graphs

Exam code: 0607

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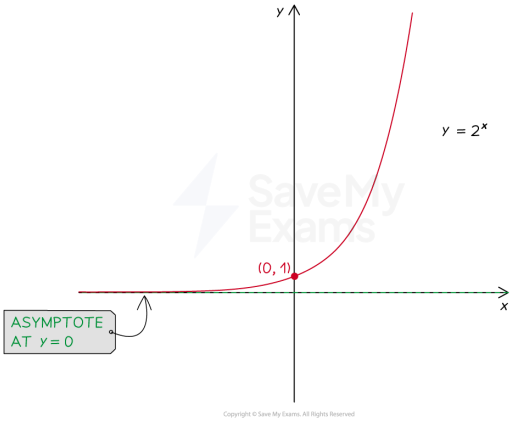
Exponential Graphs

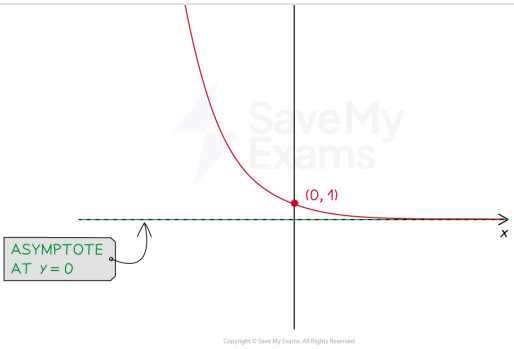
What is an exponential graph?

- An **exponential** graph is of the form $y = a^x$
  - When $a > 1$, $y = a^x$ represents **exponential growth**
    - Values of y increase as x increases
  - When $0 < a < 1$, $y = a^x$ represents **exponential decay**
    - a is positive but **less than 1**
    - Values of y decrease as x increases
- Graphs of both **exponential growth** and **exponential decay**
  - have a **horizontal asymptote** at $y = 0$
  - **do not** have a **vertical asymptote**
  - have a y-intercept of $(0, 1)$



Help me understand

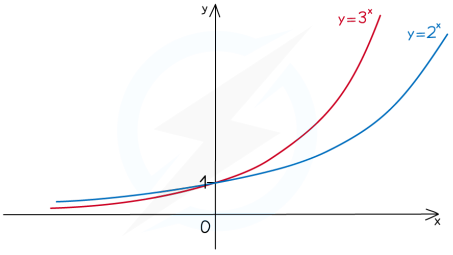
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



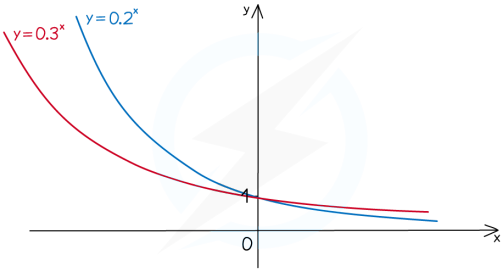
- **Exponential decay** can also be identified by a **negative power** using **index laws**
  - $\left(\frac{1}{2}\right)^x = (2^{-1})x = 2^{-x}$
  - This has the form $y = a^{-x}$ where $a > 1$

How do I compare exponential graphs?

- For $y = a^x$ where $\boldsymbol{a > 1}$:
  - The graph with a higher value of a is the "higher" graph for $\boldsymbol{x > 0}$
    - but the "lower" graph for $\boldsymbol{x < 0}$



- For $y = a^x$ where $\boldsymbol{0 < a < 1}$:
  - The graph with a higher value of a is the "higher" graph for $\boldsymbol{x > 0}$
    - but the "lower" graph for $\boldsymbol{x < 0}$



A: $y = 4^{2x}$

B: $y = \frac{6}{x}$

C: $y = 0.3^x$

D: $y = 5x^3$

A is an exponential graph as the *x* is in the power
The base number is greater than 1 so it is *exponential growth* (not decay)

B has the *x* as the denominator in a fraction
It is a reciprocal graph

C is an exponential graph as the *x* is in the power
The base number is between 0 and 1 so it is *exponential decay*

D has *x* raised to the power 3
It is a cubic graph

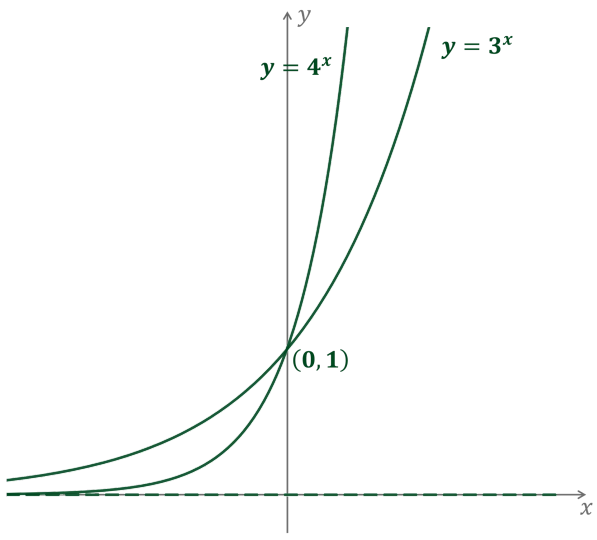
C: $y = 0.3^x$ shows exponential decay

(b) On the same set of axes, sketch the graphs of $y = 3^x$ and $y = 4^x$.

Both graphs will have the "typical" $y = a^x$ exponential growth shape for $a > 1$

$y = 4^x$ will be the "higher" graph for $x > 0$ and "lower" graph for $x < 0$

Both graphs go through $(0, 1)$ and have an asymptote along the x-axis



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